

Sampling record

2026 R1 / HP-FD — record 2026R1-HP-FD-2026R1-HP-FD

1. Identification

Programme	SaniTap safe drinking water supply, Madagascar — Gold Standard SDWS methodology v2.0 — operator MadAvance
Stratum	HP-FD — Hand pumps — Fort-Dauphin (Taolagnaro)
Round	2026 R1
Draw date/time (UTC)	2026-09-09T08:55:38.278Z
Drawn by	Adriaan Mol (SaniTap) — test draw via SaniTap Sampler CLI
Record id	2026R1-HP-FD-2026R1-HP-FD
Tool	SaniTap Sampler v1.3.0 (7a84817) — https://sanitap-water.github.io/sanitap-sampler/

2. Method

The sample is a two-stage cluster sample as described in the SaniTap Water Quality Testing Protocol v2.1, section 6.4, applied separately to each stratum.

Stage 1 — sources. The eligible sources of the stratum are listed in a fixed order (commune, then water point id) with their number of households served. Sources are drawn at random with probability proportional to households served by systematic sampling: the list is cut into as many equal intervals of households as there are sources to select, one random start is drawn in the first interval, and the source containing each successive point (start, start + interval, start + 2 x interval, ...) is selected. Ordering by commune spreads the selection across communes in proportion to their households. A source serving more households than one interval is selected with certainty and the interval is recomputed on the others, so no source is selected twice. Sources with no recorded household count receive the median count of the stratum.

Replacement list. After the main selection, replacement sources are drawn one by one from the remaining sources, again with probability proportional to households served, and listed in draw order. A replacement is used only when a selected source cannot be sampled (pump broken, inaccessible, refusal); the reason is written on the field sheet and in mWater, and replacements are taken strictly in the listed order.

Stage 2 — households. At each source the sampler counts the households served (K), numbers them clockwise from the source starting at the nearest, and draws the household numbers by the field rule: N random numbers between 1 and K plus replacements, generated from the round seed, the source id and K. A point-of-use (PoU) sample is taken from the stored drinking water of each selected household.

Point-of-collection (PoC) sample. One PoC sample is taken at the source outlet on each day households of that source are sampled, after disinfection of the spout and flushing, so that every PoU sample is paired with a PoC sample of the same source and day.

3. Sampling frame

A source is eligible when it belongs to the MadAvance water point register (mWater group group:aaaf0a14e4ce44eaa7a2bcfd1c74aa56), has at least one final water quality result in the form "Clean Water || Water Quality Testing_SDWS_3_Result" that meets the health-based rule (E. coli = 0 CFU/100 mL, arsenic <= 10 µg/L, fluoride <= 1.5 mg/L, and, where measured, nitrate <= 50 mg/L and manganese <= 0.08 mg/L; pH, conductivity, turbidity and iron are recorded but do not exclude), is not abandoned, identified-only, proposed, reported not functional in its latest maintenance record or of a non-hand-pump type, and lies in a district mapped to a stratum (Taolagnaro -> HP-FD, Maroantsetra -> HP-MA); the Marolinta area (Beloha and Amboasary districts) is excluded and any other district is unassigned.

Source of the frame	Fetches live from the mWater API by the tool
mWater group	group:aaaf0a14e4ce44eaa7a2bcfd1c74aa56
mWater forms used	7b33c5d7e5074808a94915939a5a0783, 8aa2dd78eb1f460f8f43db7935955846, 86cf66efdd3749dd8a121314bab3675a

Fetches at (UTC) 2026-09-09T08:55:38.233Z

Stratum filter at fetch HP-FD

Item	Value
Water points in the MadAvance group	908
with at least one passing SDWS 3 result	728
excluded: no passing result	180
excluded: abandoned / not functional / not a hand pump	1
excluded: Marolinta area (Beloha, Amboasary)	0
unassigned district	0
eligible in stratum HP-FD	126
eligible in stratum HP-MA	601

Eligible sources in this stratum 126 (HP-FD)

Frame file SHA-256 de62c0b1ae0b11946bcfebfbc2e03b61a1fbeb0372db41d1b8b9664574443f0e

4. Randomness and reproducibility

Seed string	2026R1-HP-FD
Seed word (xmur3, uint32)	3423278703
Generator	mulberry32 (32-bit), seeded with the xmur3 hash of the seed string — SaniTap Sampler v1.3.0 (7a84817)
Stage 1 numbers	total households 18237; interval 1519.7500; random start 934.4240; certainty selections 0; sources with imputed count 2 (value 95)

The same seed string, the same frame file and the same parameters reproduce exactly the same selection on any device. To reproduce this draw:

1. Open <https://sanitap-water.github.io/sanitap-sampler/> (any browser; the tool runs offline after the first load).
2. Data: choose "CSV file", load the frame file whose SHA-256 is de62c0b1ae0b11946bcfebfbc2e03b61a1fbeb0372db41d1b8b9664574443f0e (the file is filled with this record and is also produced by "Download loaded frame").
3. Parameters: round "2026 R1", stratum HP-FD, target 58 samples, 5 households per source, replacement fraction 20 %, method "Protocol v2.1 section 6.4 (systematic PPS by households served)", seed "2026R1-HP-FD".
4. Press "Draw the sample". The audit record shown must list the same sources in the same order as this document, and its SHA-256 must equal the value in the footer once exported.

5. Design check (CDM 90/10 rule)

Item	Value
Target PoU samples	58
Sources to select	12 (12 selected)
Households per source (m)	5
PoU samples planned (n)	60
Expected pass rate (p)	0.95
Confidence / precision	90 % / 10 % relative to p
Required sample size	15
ICC assumed	0.1
Design effect $1 + (m - 1) \times \text{ICC}$	1.40
Effective sample size n / DEFF	42.9

Item	Value
Check	PASS — effective n is at least the required n

6. Selected sources (visit in any order; numbers are draw order)

No.	Source id	Name	Commune	Households	Field rule (K, household numbers)
1	742895168	IndiaMark 131	Ampasy Nahampoana	386	K = ____ (numbers generated in the tool when K is entered)
2	742895120	Canzee 127	Analapatsy	805	K = ____ (numbers generated in the tool when K is entered)
3	742894772	IndiaMark 94	Ankaramena	832	K = ____ (numbers generated in the tool when K is entered)
4	742893805	Canzee 1	Fort-Dauphin	447	K = ____ (numbers generated in the tool when K is entered)
5	742894442	IndiaMark 61	Ifarantsa	33	K = ____ (numbers generated in the tool when K is entered)
6	742895182	IndiaMark 133	Mahatalaky	82	K = ____ (numbers generated in the tool when K is entered)
7	742894552	IndiaMark 72	Manambaro	67	K = ____ (numbers generated in the tool when K is entered)
8	742894662	Canzee	Manambaro	765	K = ____ (numbers generated in the tool when K is entered)
9	742895814	Canzee 158	Manantenina	228	K = ____ (numbers generated in the tool when K is entered)
10	742894820	IndiaMark	Ranopiso	238	K = ____ (numbers generated in the tool when K is entered)
11	742894930	IndiaMark 110	Ranopiso	273	K = ____ (numbers generated in the tool when K is entered)
12	742894150	Canzee 31	Soanierana	502	K = ____ (numbers generated in the tool when K is entered)

7. Replacement sources (use strictly in this order)

Use a replacement only when a selected source cannot be sampled; record the reason.

No.	Source id	Name	Commune	Households	Field rule (K, household numbers)
R1	742895742	Canzee 156	Manantenina	232	K = ____ (numbers generated in the tool when K is entered)
R2	742894693	IndiaMark 86	Soanierana	120	K = ____ (numbers generated in the tool when K is entered)
R3	670401763	Canzee	Mahatalaky	168	K = ____ (numbers generated in the tool when K is entered)

8. Field rule

At each source: count all households that draw water from it (K), number them clockwise from the source starting at the nearest, enter K in the tool to obtain 5 household numbers plus 2 replacement numbers, and sample those households. The numbers are generated from the seed, the source id and K, so they can be re-generated by the verifier.

9. Notes generated by the tool

- {"code": "imputed_weights", "n": 2, "value": 95}